

Document 2: Pedalboard Layout & Row Ownership Specification

2.1 Topology Mapping

The pedalboard is organized in a strict 2 Rows × 3 Columns grid topology, flattened sequentially into 1D linear storage using a row-major pattern:

- Slot 0: Row 0, Column 0 (Top-Left)
- Slot 1: Row 0, Column 1 (Top-Middle)
- Slot 2: Row 0, Column 2 (Top-Right)
- Slot 3: Row 1, Column 0 (Bottom-Left)
- Slot 4: Row 1, Column 1 (Bottom-Middle)
- Slot 5: Row 1, Column 2 (Bottom-Right)

2.2 Dynamic Canvas Partitioning Engine

The 128 vertical rows are divided exclusively among unbypassed, active pedals. Unpopulated slots receive 0 rows and are omitted from allocation prior to drawing. Let N represent the count of active, populated pedals ($1 \leq N \leq 6$).

- **Base Slices:** Each active pedal receives a base block allocation of $S_{base} = \lfloor 128 / N \rfloor$ rows.
- **Remainder Distribution:** Remainder rows ($R = 128 \bmod N$) are distributed one-by-one to the first R active pedals in sequential index order. Maximum row imbalance is capped at 1 row.

2.3 State-Machine Lock Mechanics

- **UNLOCKED State:** Active when the canvas framebuffer is empty. Users can change pedal types or alter slot assignments. Any modification triggers a recalculation of row ranges.
- **LOCKED State:** Triggered on the first pixel write. Pedal routing choices and slot occupancy are frozen. Hot-swapping is blocked while active data populates the canvas.
- **State Reset:** Returns to UNLOCKED when the user clears the canvas, executes an undo operation resulting in an empty canvas, or removes the plugin instance from the parent DAW chain.